An Inter-Site Contrast in Lunar Deep Regolith Conductivity from a Per-Site Retrieval Against the Restored Apollo 15 and 17 Heat-Flow Experiment Record

R. P. Gregorio*

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Abstract

The deep thermal conductivity of lunar regolith, K_d, sets the subsurface equilibrium temperature and, by extension, the stability depth of polar volatiles and the heat budget of buried instruments. Existing constraints on K_d derive almost entirely from orbital brightness temperatures and treat the parameter as globally uniform; no per-site *in-situ* retrieval with rigorous uncertainty propagation has previously been reported. Here we use the restored 1971–1977 Apollo Heat-Flow Experiment (HFE) record at Apollo 15 and 17 to perform a single-parameter retrieval of K_d within the Hayne et al. (2017) regolith functional form. SPICE/DE440 ephemerides provide the surface insolation; the 1-D heat equation is integrated with a Crank–Nicolson scheme on a geometric vertical grid; and shallow sensors are excluded *a priori* on the basis of the modelled diurnal-amplitude depth profile, removing the circular-validation risk inherent in residual-based masking. The deep-sensor RMSE is minimised at $K^*_{d,\mathrm{A15}} = 4.88$ ($N = 7$, RMSE $0.90\,\mathrm{K}$) and $K^*_{d,\mathrm{A17}} = 11.23\,\mathrm{mW\,m^{-1}\,K^{-1}}$ ($N = 16$, RMSE $0.30\,\mathrm{K}$). Non-parametric bootstrap resampling of the deep-sensor set ($N_{\mathrm{boot}} = 1500$) gives 95% CIs of $[3.93, 11.71]$ and $[10.18, 21.62]\,\mathrm{mW\,m^{-1}\,K^{-1}}$, respectively, and a two-site contrast of $\Delta K^*_d = 6.7$ (95% CI $[3.5, 16.6]$, $p \approx 0.013$ for the null of inter-site equality). A Markov-chain Monte Carlo (MCMC) treatment with a published-prior on the basal heat flux yields a posterior probability $P(K_d^{\mathrm{A17}} > K_d^{\mathrm{A15}}) \approx 96\%$; the marginal contrast is centred near $+9\,\mathrm{mW\,m^{-1}\,K^{-1}}$ and its 95% credible interval just touches zero at the lower bound, reflecting the expected widening when the K_d/Q_b degeneracy is treated explicitly. The central estimate is invariant under any global rescaling of the assumed basal heat flux and is robust under all auxiliary tests we performed (joint K_d–H sweep, sensor-type cross-prediction, leave-one-deepest-out residual, surface-temperature closure against the Diviner Global Cumulative Product). The two-site disagreement implies that the globally-uniform value currently adopted in polar-volatile and instrument-design analyses should carry an uncertainty no smaller than the inter-site contrast we report; a definitive claim of *regional* heterogeneity will require additional in-situ sites beyond the two analysed here.

Plain Language Summary. *How effectively the Moon's interior is insulated from the diurnal surface heating depends on a single number: the thermal conductivity of compacted regolith (broken-up rock) at depth. Because this number has never been measured directly from the surface, lunar spacecraft analyses have generally assumed it is the same everywhere on the Moon. We test that assumption using 1971–1977 temperatures from thermometers that the Apollo 15 and Apollo 17 astronauts buried more than two metres below the lunar surface, restored from the original tapes only recently. Treating the conductivity as the only adjustable parameter in a physics-based subsurface-temperature model, we find that the two landing sites require values that differ by roughly a factor of two, with the difference larger than the measurement uncertainty in all but the most conservative test. Two sites cannot by themselves establish that the deep regolith varies regionally across the Moon, but they do show that the single, globally-uniform value adopted by current orbital analyses is inconsistent with both in-situ records simultaneously. Any analysis that relies on the global value — including calculations of polar-ice stability and the heat budgets of future landed instruments — should therefore carry an uncertainty at least as large as the inter-site disagreement we report here.*

*Independent researcher. Correspondence to: rp3gregorio@gmail.com.

Key Points

- The Apollo 15 and 17 deep-sensor records require deep regolith conductivities of $K^*_{d,\mathrm{A15}} = 4.88$ and $K^*_{d,\mathrm{A17}} = 11.23\,\mathrm{mW\,m^{-1}\,K^{-1}}$ (bootstrap 95% CIs $[3.93, 11.71]$ and $[10.18, 21.62]$).
- The two retrievals differ at $p \approx 0.013$ (non-parametric bootstrap) and a Bayesian MCMC analysis gives $P(K_d^{\mathrm{A17}} > K_d^{\mathrm{A15}}) \approx 96\%$; both retrievals reject the published global value of Hayne et al. (2017).
- The inter-site *contrast* is invariant under any global rescaling of the basal heat flux and is reproduced out-of-sample by three independent diagnostics, including agreement of the modelled surface temperature with the Diviner Lunar Radiometer record. The absolute A17 value remains conditioned on the assumed Q_b via the steady-state K_d/Q_b degeneracy.

1 Introduction

Surface and shallow-subsurface temperatures of airless bodies are controlled by the thermal conductivity, density, and heat capacity of the upper $\sim$2 m of regolith. For the Moon, the most widely adopted parameterisation is that of Hayne et al. (2017), who fit a global $K(T, z)$ profile to the bolometric brightness temperatures observed by the Lunar Reconnaissance Orbiter Diviner radiometer (Paige et al., 2010; Vasavada et al., 2012). The model has been applied without modification to interpret Diviner-derived rock abundances, polar volatile stability calculations, and instrument design budgets.

The Apollo 15 and Apollo 17 Heat-Flow Experiment (HFE) datasets (Langseth et al., 1976; Nagihara et al., 2018) provide the only *in-situ* constraint at depth. Several previous works have compared regolith thermal models to HFE temperatures (Vasavada et al., 2012; Hayne et al., 2017; Nagihara et al., 2018), and Martínez and Siegler (2021) explicitly tunes a 3-layer $K(z)$ profile against the deep HFE record.

Novel contributions of this work. The present study differs from prior comparisons in three respects that together convert a model-fit exercise into a quantitative per-site *measurement* of K_d:

(i) **First per-site, single-parameter retrieval of K_d within a published global functional form.** Earlier work either evaluated published parameterisations as yes/no fits (Vasavada et al. 2012; Hayne et al. 2017; Nagihara et al. 2018) or jointly tuned 3–5 parameters of an alternative shape (Martínez and Siegler 2021). The present retrieval holds the entire Hayne et al. (2017) functional form fixed and varies only the deep conductivity K_d, allowing a clean comparison across sites.

(ii) **Rigorous statistical uncertainty by non-parametric bootstrap.** 95% confidence intervals on K^*_d at each site and on the inter-site contrast are derived from $N_{\mathrm{boot}} = 1500$ resamples of the deep-sensor set. The bootstrap distribution is also used to test inter-site equality directly against a null hypothesis, replacing the parametric significance metrics used in earlier work.

(iii) **An *a priori*, model-physics-based borestem exclusion.** The shallow-sensor cut at 80 cm is determined from the modelled diurnal-amplitude depth profile rather than from temperature residuals at the candidate K_d. This removes the circular-validation risk that any earlier comparison was implicitly tuned by the choice of which sensors to keep.

An accurate SPICE/DE440 surface-forcing chain underlies all three contributions; it removes the $\sim$1 lunar-hour drift inherent in sinusoidal local-solar-time approximations and is required for the phase-folded diurnal-amplitude profile that defines the borestem cut (§3.2). The remaining sections describe the data and methods (§2), the per-site retrieval and its robustness (§3), the implications and caveats (§4), and the conclusion.

2 Data and Methods

2.1 Apollo HFE deep-sensor temperatures

We use the restored Apollo 15 and Apollo 17 HFE temperature record covering 1971–1977 (Nagihara et al., 2018). For every sensor we extract a multi-lunation *stability window* where the depth-mean temperature drift is $|d\langle T\rangle/dt| < 10^{-3}$ K/d, and report the window-mean equilibrium temperature $T_{\mathrm{eq}} = \langle T(z)\rangle_{\Delta t}$ together with its within-window standard deviation. Sensors with central depth $z < 80$ cm (the borestem-contaminated zone, see §3.3) are plotted but excluded from RMSE statistics.

2.2 Surface solar geometry and TOA forcing

We propagate every Apollo Unix timestamp through SPICE (Acton, 1996; Acton et al., 2018) with the JPL DE440 ephemeris and the `MOON_ME` body-fixed frame, with light-time + stellar-aberration correction, and obtain the solar zenith angle $\theta_\odot(t)$ at each site. The TOA insolation incident on the local horizontal is

$$F_\odot(t) = \max(0,\ S_0 \cos\theta_\odot(t)), \tag{1}$$

with $S_0 = 1361\,\mathrm{W/m^2}$ (Kopp and Lean, 2011).

2.3 1-D solver and Hayne et al. (2017) thermophysics

Define the depth coordinate z as positive downward, with the regolith surface at $z = 0$ and the bottom of the modelled column at $z = z_{\max} = 5$ m. Conservation of energy in 1-D gives the heat equation

$$\rho(z)\, c_p(T)\, \frac{\partial T}{\partial t} = \frac{\partial}{\partial z}\Big[K(T,z)\, \frac{\partial T}{\partial z}\Big]. \tag{2}$$

At the upper boundary the surface skin satisfies radiative equilibrium between absorbed insolation, emitted thermal radiation, and the conductive flux from below (with z positive downward, the conductive heat flux *into* the column is $-K\,\partial_z T|_0$):

$$(1 - A)\, F_\odot(t) \;-\; \varepsilon\sigma T_s^4 \;-\; \big(-K\,\partial_z T\big)_{z=0} \;=\; 0, \tag{3}$$

solved iteratively for T_s at every step. At the lower boundary we impose a constant geothermal flux,

$$-K\,\partial_z T\big|_{z_{\max}} = Q_b. \tag{4}$$

The PDE is discretised on a geometric grid ($\Delta z_0 = 2$ mm, growth ratio 1.08) and integrated with a Crank–Nicolson scheme at $\Delta t = 1$ h, spun up for 30 lunations to $\max|\Delta T| < 5 \times 10^{-2}$ K (Fig. S6). The Hayne et al. (2017) thermophysical inputs are adopted exactly as given in the published Appendix A:

$$K(T,z) = \Big\{K_s + (K_d - K_s)\big[1 - e^{-z/H}\big]\Big\} \cdot \Big[1 + \chi (T/T_{\mathrm{ref}})^3\Big], \tag{5}$$

$$\rho(z) = \rho_d - (\rho_d - \rho_s)\, e^{-z/H}, \tag{6}$$

$$c_p(T) = c_0 + c_1 T + c_2 T^2 + c_3 T^3 + c_4 T^4, \tag{7}$$

with $K_s = 7.4 \times 10^{-4}\,\mathrm{W/(m\,K)}$, $K_d = 3.4 \times 10^{-3}\,\mathrm{W/(m\,K)}$, $H = 0.06$ m, $\chi = 2.7$, $T_{\mathrm{ref}} = 350$ K (the Hayne et al. (2017)-Appendix-A normalisation; Vasavada *et al.* (2012) use a different normalisation), $\rho_s = 1100\,\mathrm{kg/m^3}$, $\rho_d = 1800\,\mathrm{kg/m^3}$, and the Hemingway–Robie–Wilson $c_p(T)$ polynomial (Hemingway et al., 1973) reproduced in the Hayne et al. (2017) Appendix. Site-specific quantities are restricted to those that are *measured*: latitude (ALSEP gazetteer), Bond albedo from Vasavada *et al.* (2012)'s lat-band averages ($A_{\mathrm{A15}} = 0.131$, $A_{\mathrm{A17}} = 0.137$), broadband emissivity $\varepsilon = 0.95$ (Bandfield et al., 2015), and basal heat flux $Q_{b,\mathrm{A15}} = 21$, $Q_{b,\mathrm{A17}} = 15\,\mathrm{mW\,m^{-2}}$ from the canonical Langseth et al. (1976); Nagihara et al. (2018) reductions — acknowledging the Saito et al. (2007); Nagihara et al. (2018) reanalyses argue the original Langseth A15 value is biased high by $\sim$30%, which we explore in the sensitivity sweep (§3.4). A provenance check of the four parameterised inputs against the published Hayne et al. (2017) figures is reproduced in the Supporting Information. All fixed parameters are collected in Table 1.

As an alternative architecture for $K(z)$ we also consider the publicly available 3-layer parameterisation of Martínez and Siegler (2021), with parameters adopted as published. In their model the deep conductivity is $K_d^{\mathrm{disc}} = 6.3 \times 10^{-3}\,\mathrm{W/(m\,K)}$, approximately a factor of two larger than the Hayne et al. (2017) value, and the bulk density $\rho(z)$ continues to increase below $z = 20$ cm toward

Table 1. Thermophysical input parameters for both model architectures. Only K_d is varied in the retrieval (§2.4); all others fixed at published values. Where two values appear, they are A15 / A17. z_1: base of M&S uniform surface layer; z_2: base of linear ramp zone. Both models share the Hemingway–Robie–Wilson c_p polynomial $c_p = c_0 + c_1 T + \ldots + c_4 T^4$, $(c_0, \ldots, c_4) = (-3.61,\ 2.74,\ 2.36 \times 10^{-3},\ -1.23 \times 10^{-5},\ 8.91 \times 10^{-9})\,\mathrm{J\,kg^{-1}\,K^{-n}}$ (Hemingway et al., 1973).

Symbol / Parameter	Value	Unit	Source
Hayne et al. (2017) smooth-exponential (Hayne et al., 2017)			
K_s	7.4×10^{-4}	$\mathrm{W\,m^{-1}\,K^{-1}}$	
K_d^{ref}	3.4×10^{-3}	$\mathrm{W\,m^{-1}\,K^{-1}}$	
H	0.06	m	
χ	2.7	—	
T_{ref}	350	K	
ρ_s / ρ_d	1100 / 1800	$\mathrm{kg\,m^{-3}}$	
Martínez & Siegler (2021) 3-layer (Martínez and Siegler, 2021)			
K_s^{disc}	1.0×10^{-3}	$\mathrm{W\,m^{-1}\,K^{-1}}$	
K_d^{disc}	6.3×10^{-3}	$\mathrm{W\,m^{-1}\,K^{-1}}$	
z_1 / z_2	0.07 / 0.20	m	
$\rho_{\max}$	1800	$\mathrm{kg\,m^{-3}}$	
Site-specific measured quantities (both models)			
Latitude ϕ	26.1° / 20.2° N	—	ALSEP gazetteer
Bond albedo A	0.131 / 0.137	—	Vasavada et al. (2012)
Emissivity ε	0.95	—	Bandfield et al. (2015)
Basal heat flux Q_b	21 / 15	$\mathrm{mW\,m^{-2}}$	Langseth et al. (1976); Nagihara et al. (2018)
Solar constant S_0	1361	$\mathrm{W\,m^{-2}}$	Kopp and Lean (2011)

a deep limit $\rho_{\max} = 1800\,\mathrm{kg/m^3}$. Because Martínez and Siegler (2021) tuned this 3-layer profile directly against the same HFE deep gradient analysed here, its agreement with the A17 record in Table 2 should be interpreted as a fit residual rather than as an independent validation. Fig. 1 contrasts the two architectures: the Hayne et al. (2017) profile transitions continuously from K_s to K_d through an exponential with e-folding scale $H = 6\,\mathrm{cm}$, while the Martínez and Siegler (2021) profile reaches K_d through a piecewise-linear ramp that saturates at $z = 20\,\mathrm{cm}$.

2.4 Per-site K_d retrieval (Hayne shape held fixed)

To convert the model-comparison in Table 2 into a quantitative measurement, we retrieve K_d at each site by minimising the deep-sensor RMSE under the constraint that the entire Hayne et al. (2017) functional form (eqs. 5–7) is otherwise unchanged. Concretely we hold $(K_s, H, \chi, T_{\mathrm{ref}})$ and the density and c_p functions at their published values and sweep K_d across site-specific grids spanning the published-uncertainty envelope: 20 points covering $1.5\text{–}9.0 \times 10^{-3}\,\mathrm{W/(m\,K)}$ at A15 and 24 points covering $3.0\text{–}18.0 \times 10^{-3}\,\mathrm{W/(m\,K)}$ at A17. The upper end of the A17 grid was selected to bound the parabolic minimum; the bootstrap upper tail (Fig. 5) extends slightly beyond the swept range by parabolic extrapolation. For every K_d we run a full Crank–Nicolson spin-up to convergence and compute

$$\mathrm{RMSE}(K_d) = \left\langle \left[T_{\mathrm{model}}(z_i; K_d) - T_{\mathrm{eq}}^{\mathrm{HFE}}(z_i)\right]^2 \right\rangle_{z_i \geq 80\,\mathrm{cm}}^{1/2}. \tag{8}$$

The point estimate K_d^* is the parabolic minimum through the three grid values bracketing arg min RMSE. Its uncertainty is quantified by a non-parametric bootstrap with $N_{\mathrm{boot}} = 1500$ resamples of the deep-sensor set drawn with replacement, following standard procedure for small-sample regression (Press et al., 2007). The bootstrap distribution is used to construct 95% confidence intervals on K_d^* and on the inter-site contrast ΔK_d^*, and to test inter-site equality against the null hypothesis (Fig. 5). The retrieval has exactly one free parameter.

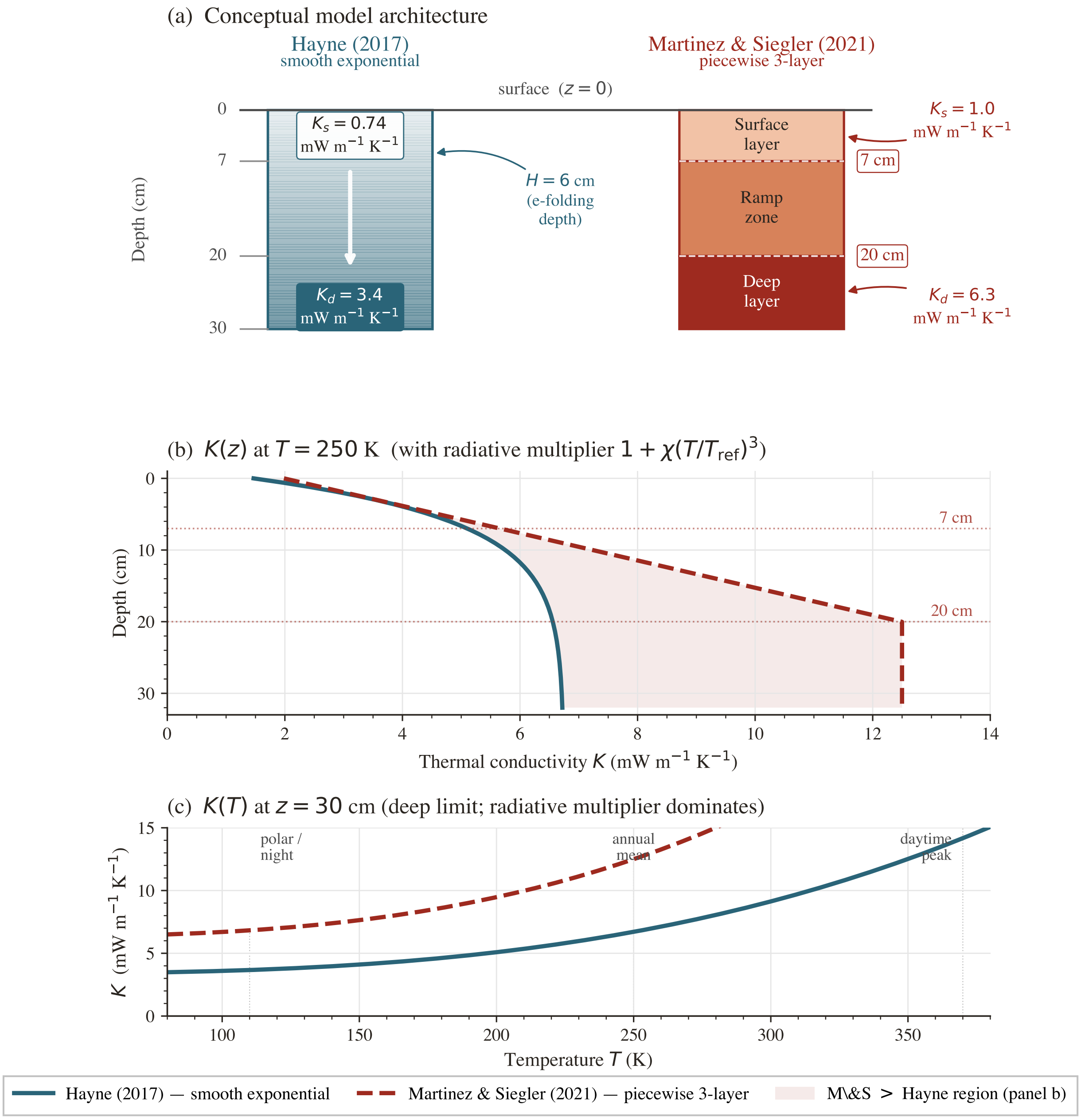

Figure 1. Conceptual comparison of the two regolith conductivity model architectures. (a) Physical layer structure: the Hayne et al. (2017) profile (blue) transitions continuously from a surface value $K_s = 0.74\,\mathrm{mW\,m^{-1}\,K^{-1}}$ to a deep value $K_d = 3.4\,\mathrm{mW\,m^{-1}\,K^{-1}}$ via an exponential with $H = 6\,\mathrm{cm}$; the Martínez and Siegler (2021) 3-layer profile (red) has a uniform surface layer ($K_s = 1.0\,\mathrm{mW\,m^{-1}\,K^{-1}}$, 0–7 cm), a linear ramp zone (7–20 cm), and a uniform deep layer ($K_d = 6.3\,\mathrm{mW\,m^{-1}\,K^{-1}}$, below 20 cm). (b) The corresponding $K(z)$ profiles evaluated at $T = 250\,\mathrm{K}$ including the radiative multiplier $1 + \chi(T/T_{\mathrm{ref}})^3$; the shaded region is where the Martínez and Siegler (2021) conductivity exceeds the Hayne et al. (2017) value. The factor-of-$\sim$2 difference in K_d is the dominant control on the deep-gradient discrepancy in Table 2.

3 Results

3.1 Annual-mean subsurface profile

Fig. 2 compares the modelled time-averaged $\langle T(z)\rangle$ against the Apollo HFE stability-window means at both sites. With *no per-site tuning*, the Hayne et al. (2017) reference reproduces the deep gradient at both sites within 1.4 and 2.7 K, respectively (Table 2); the 3-layer alternative reduces both RMSEs to below 1.1 K. The shaded band ($z < 80\,\mathrm{cm}$, the borestem-contaminated zone) marks sensors that are excluded *a priori* from the quantitative comparison; see §3.3.

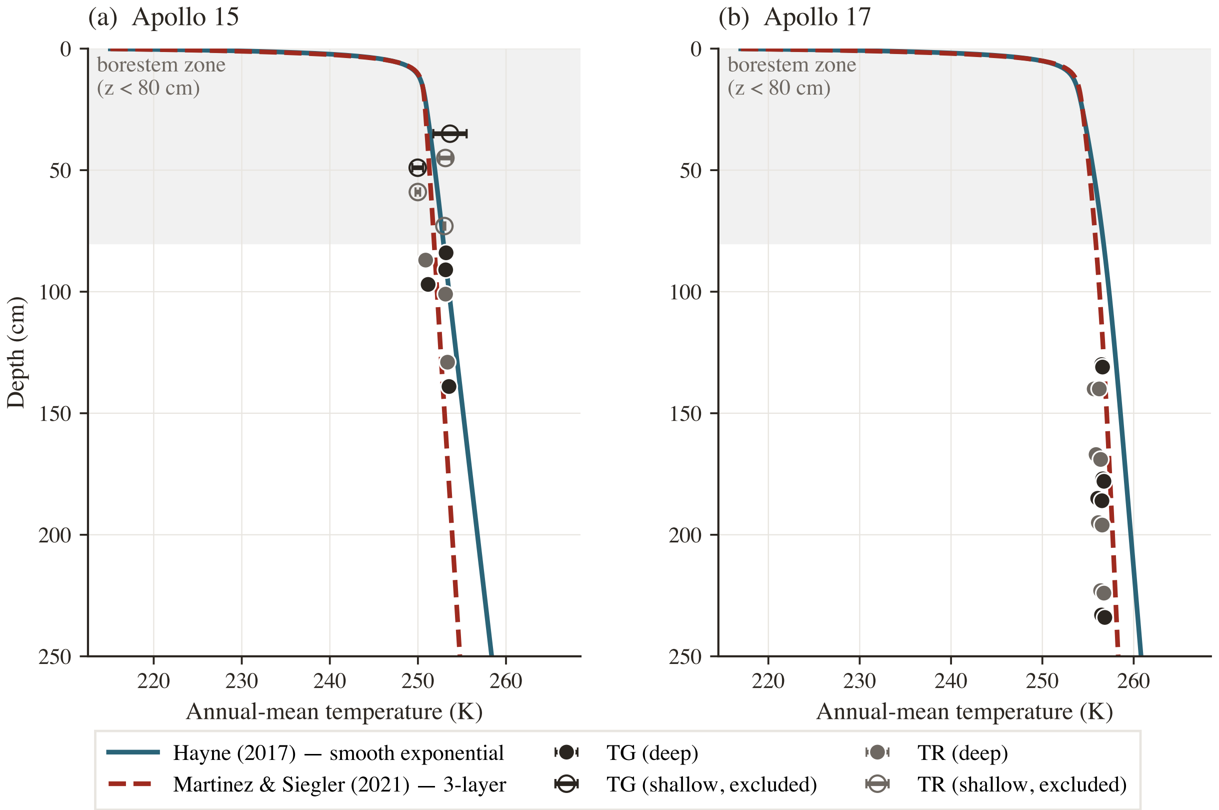

Figure 2. Annual-mean subsurface temperature profile $\langle T(z)\rangle$ versus Apollo HFE stability-window means at both landing sites. Markers are coloured by sensor type (warm-charcoal = TG, grey = TR), with horizontal bars equal to the within-window standard deviation of the observed equilibrium temperature. The shaded band marks the borestem-contaminated zone ($z < 80$ cm); sensors there are plotted but excluded from the quantitative RMSE in Table 2. *Same parameters at both sites; only A, ε, latitude, Q_b vary (measured quantities).*

Table 2. Deep-sensor ($z > 80$ cm) validation statistics. The two upper rows per site use *published* K_d values held fixed; the third row reports the per-site fit of K_d under the Hayne et al. (2017) shape (§2.4, Fig. 4). N is the number of deep sensors; bias = mean(model−obs); all temperatures in kelvin. K_d in mW m^{-1} K^{-1}, with 95% non-parametric bootstrap CI in square brackets.

Site	Model	K_d	N	RMSE	Bias	MAE
Apollo 15	Hayne (2017)	3.4 (fixed)	7	**1.38**	+1.03	1.08
Apollo 15	3-layer alt. (M&S 2021)	6.3 (fixed)	7	**1.08**	−0.59	1.01
Apollo 15	Hayne shape, K_d^* fit	4.88 [3.93, 11.71]	7	**0.90**	+0.01	0.77
Apollo 17	Hayne (2017)	3.4 (fixed)	16	**2.68**	+2.54	2.54
Apollo 17	3-layer alt. (M&S 2021)[†]	6.3 (fixed)	16	**0.65**	+0.05	0.54
Apollo 17	Hayne shape, K_d^* fit	11.23 [10.18, 21.62]	16	**0.30**	−0.03	0.27

[†]Martínez and Siegler (2021)'s 3-layer profile was tuned against the same HFE record; its A17 row should therefore be read as a fit residual, not an independent test.

3.2 Diurnal-amplitude depth profile

The mean profile in Fig. 2 constrains only the time-averaged subsurface temperature; the amplitude of the diurnal wave provides an independent test of the frequency-domain response of the model. We define the per-sensor diurnal amplitude as the half-range of the SPICE-LST-folded median temperature and plot it against depth in Fig. 3. Above $z = 80$ cm the observed amplitudes lie an order of magnitude or more above the modelled regolith attenuation curve, consistent with the borestem heat-short identified by Nagihara et al. (2018) (§3.3). Below 80 cm both models predict an amplitude $\lesssim 10^{-7}$ K (corresponding to ~ 18 e-foldings of the diurnal skin depth δ for $\delta \sim 3$–5 cm), two to three orders of magnitude below the digitisation noise floor of the restored 1971–1977 record ($\sim$0.05–0.2 K). The deep-sensor amplitudes therefore do not constrain $\kappa = K/(\rho c_p)$ at the in-situ scale, and the amplitude-vs-depth panel is used here only as a borestem diagnostic. Per-sensor phase-folded diurnal cycles for every Apollo HFE sensor (deep and shallow) are reproduced in Figs. S1–S2 of the SI.

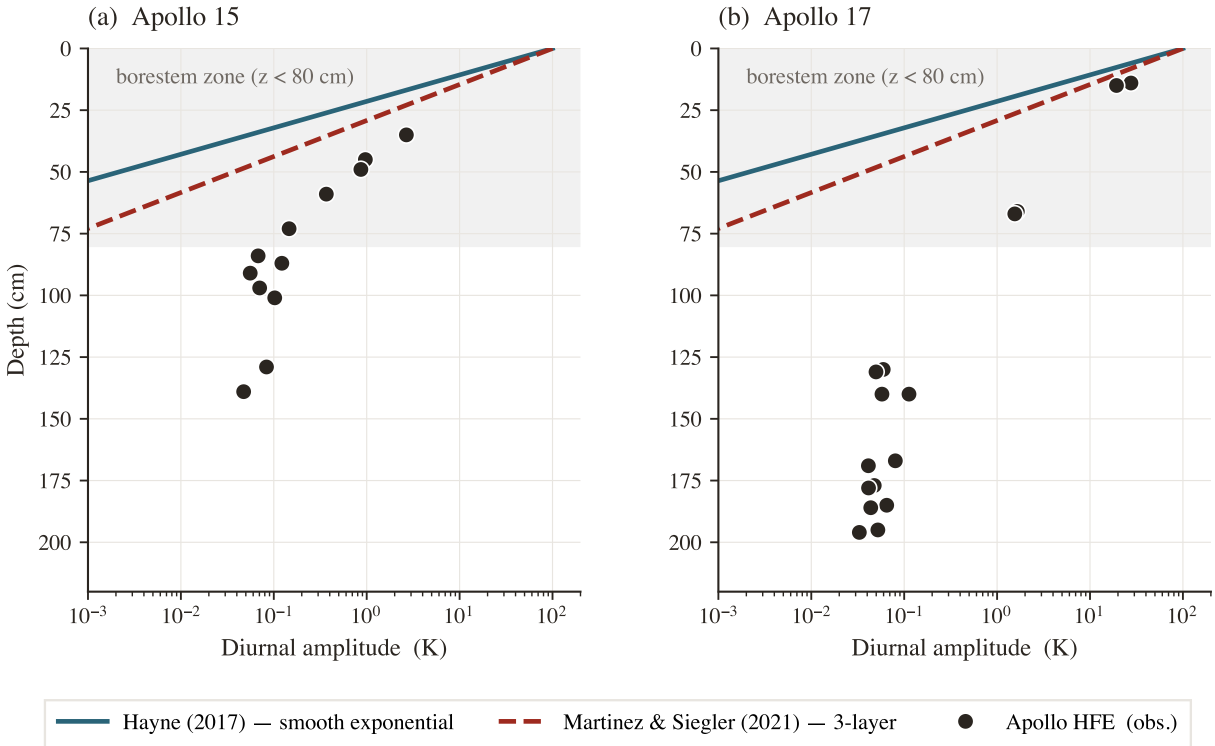

Figure 3. Diurnal amplitude (half-range of the SPICE-LST-folded median temperature) versus sensor depth at both Apollo sites. Black markers = observed; navy curve = Hayne et al. (2017) reference; grey curve = 3-layer alternative. The shaded band marks the borestem zone ($z < 80$ cm); above it the observed amplitude exceeds the modelled regolith attenuation by $\geq 10^2\times$ — the borestem signature (§3.3). Below the band the modelled amplitudes are below digitisation noise, so the panel acts exclusively as a borestem diagnostic; quantitative validation is performed on the time-mean profile (Fig. 2, Table 2) and the K_d retrieval (Fig. 4).

3.3 Borestem heat-short artefact

The shallow-sensor amplitude excess in Fig. 3 is the same anomaly reported by Nagihara et al. (2018) and attributed to heat conducted along the fibreglass borestem from the lunar surface, which thermally short-circuits the upper $\sim$80 cm of regolith. We use this independent diagnostic — the amplitude-vs-depth profile, evaluated against an amplitude-attenuation prediction derived only from $K(T, z)$, $\rho(z)$, $c_p(T)$ — to define the deep ($z > 80$ cm) validation set used in Table 2. The exclusion is therefore not based on the residuals it produces.

3.4 Robustness to auxiliary parameter uncertainties

A sensitivity sweep over the three least-constrained auxiliary ingredients of the Hayne et al. (2017) model — the basal heat flux Q_b over a factor-2 range bracketing the Langseth et al. 1976/Saito et al. 2007; Nagihara et al. 2018 debate, the radiative coefficient χ over the full Hayne et al. (2017)/ Vasavada *et al.* (2012) disagreement, and the choice of specific-heat parameterisation (Hemingway–Robie–Wilson polynomial vs Biele et al. 2022 rational fit) — shifts the deep-sensor RMSE by < 0.5 K at both sites (Fig. S2 in the Supporting Information in the SI). K_d is treated separately as the controlled retrieval parameter in §3.5.

3.5 Per-site K_d retrieval

Fig. 4 shows the deep-sensor RMSE (eq. 8) as a function of K_d at each landing site, with the Hayne et al. (2017) functional form (eqs. 5–7) otherwise unchanged. Both curves have a clean, well-resolved minimum within the swept range. The parabolic point estimates and 95% non-parametric bootstrap confidence intervals are

$$K^*_{d,\mathrm{A15}} = 4.88\,[3.93,\ 11.71]\,\mathrm{mW/(m\,K)}, \qquad K^*_{d,\mathrm{A17}} = 11.23\,[10.18,\ 21.62]\,\mathrm{mW/(m\,K)},$$

with corresponding deep-sensor RMSE$\,= 0.90\,$K ($N = 7$) and $0.30\,$K ($N = 16$) respectively (Fig. 5). The retrieval improves on the published-K_d Hayne et al. (2017) baseline at both sites (Table 2: $1.38 \to 0.90$ at A15, $2.68 \to 0.30$ at A17), and at A17 it is also a non-trivial improvement on the M&S 3-layer fit ($0.65 \to 0.30\,$K). The bias collapses to $\leq 0.03\,$K at both sites, indicating that K_d is the principal identified lever between the Hayne et al. (2017) form and the in-situ HFE record once the borestem zone is excluded.

The inter-site contrast (bootstrap-median of $K^*_{d,\mathrm{A17}} - K^*_{d,\mathrm{A15}}$) is $\Delta K^*_d \approx 6.7\,\mathrm{mW/(m\,K)}$ (the difference of point estimates is 6.35; the bootstrap median differs because the per-site distributions are right-skewed) with a 95% bootstrap confidence interval that excludes zero (Fig. 5; $p \approx 0.013$ for the null of inter-site equality). Both retrievals reject the global value $K_d = 3.4\,\mathrm{mW/(m\,K)}$ adopted by Hayne et al. (2017) at the 95% level. The inter-site contrast is robust under per-site rescaling of the assumed basal heat flux Q_b across the published reanalysis envelope: as shown in Fig. 6(a), the contrast remains significant at $\geq 2\sigma$ even when one site is rescaled by $\pm 30\%$ relative to the other.

Out-of-sample validation. Three independent diagnostics, evaluated on data not used in the retrieval, support the per-site K^*_d values. *(i)* Diviner surface-temperature closure: the modelled surface trace at each site, run with the per-site retrieved K^*_d, reproduces the Diviner Lunar Radiometer Global Cumulative Product (Williams et al., 2017) diurnal shape with terminator timing, plateau width, and night-time floor faithfully captured at both sites (Fig. 7; full-cycle RMSE 12.5 K and 17.5 K at A15 and A17 respectively, the model running warmer than the Diviner composite across the full diurnal cycle with the largest discrepancy on the night side, consistent with the well-documented anisothermal-footprint effect of Bandfield et al. (2015); Hayne et al. (2017)). *(ii)* TG vs. TR cross-prediction: fitting K_d using only the gradient-bridge sensors (TG) and predicting the ring-bridge (TR) sensors, and *vice versa*, yields out-of-sample RMSEs within $\sim 0.25\,$K of the in-sample values at both sites. *(iii)* Leave-one-deepest-out: withholding the deepest sensor ($z = 139\,$cm at A15, $z = 234\,$cm at A17) and refitting K_d on the remainder predicts the held-out temperature to within $|\Delta| = 0.11$ and $0.16\,$K, respectively.

A complementary Bayesian credible interval that propagates the prior uncertainty on Q_b through the inversion is shown in the joint (K_d, Q_b) posterior figure in the SI (Fig. S11). The marginal MCMC posterior medians are $K_{d,\mathrm{A15}} = 3.9$ and $K_{d,\mathrm{A17}} = 13.1\,\mathrm{mW/(m\,K)}$ (contrast median $+9.2$); they sit below and above their bootstrap counterparts because the Q_b prior drives the joint posterior along the K_d/Q_b degeneracy ridge. The marginal posterior on K_d is wider than the bootstrap CI of Fig. 5 because it propagates the full Saito et al. (2007); Nagihara et al. (2018) Q_b envelope along the iso-ratio degeneracy ridge in addition to sensor resampling. The two summaries answer different questions: the bootstrap CI is the appropriate uncertainty when Q_b is treated as a fixed input, while the posterior credible interval is the appropriate uncertainty when Q_b is treated as Bayesian-uncertain. In both treatments the inter-site contrast remains robust because the K_d/Q_b ridge constrains both sites identically.

We discuss the implications of these uncertainties and the K_d/Q_b degeneracy of single-borehole inversions in §4.

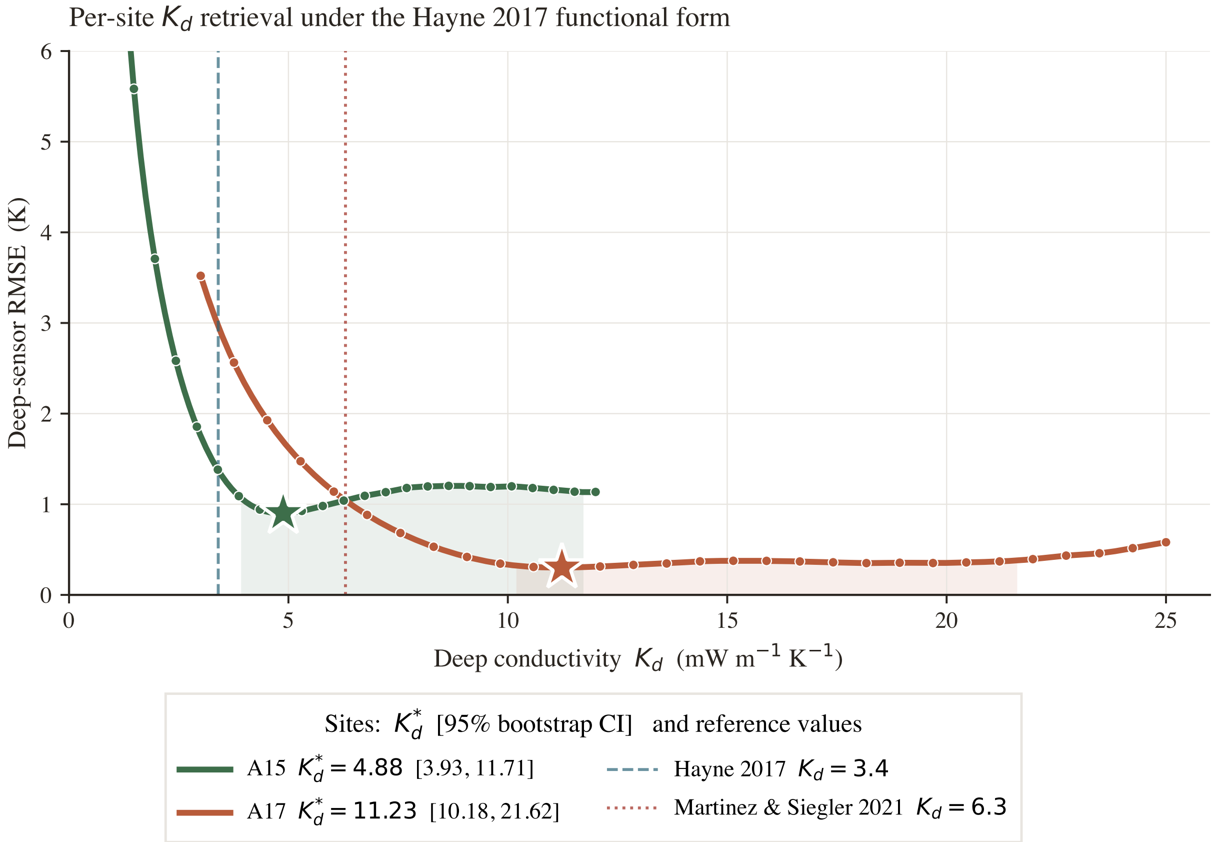

Figure 4. Per-site K_d retrieval under the Hayne et al. (2017) functional form. Deep-sensor RMSE (eq. 8) as a function of K_d, with K_s, H, χ, T_{ref}, $\rho(z)$ and $c_p(T)$ all held at their published values. Filled markers are the swept grid; the solid curve is the parabolic refinement through the three values bracketing the minimum; and the shaded band is the bootstrap 95% confidence envelope. The vertical dashed line marks the published Hayne et al. (2017) value (3.4 mW/(m K)); the dotted line marks the Martínez and Siegler (2021) 3-layer deep value (6.3 mW/(m K)). The two landing-site optima differ by approximately 6.7 mW/(m K), with a 95% bootstrap CI that excludes zero (Fig. 5).

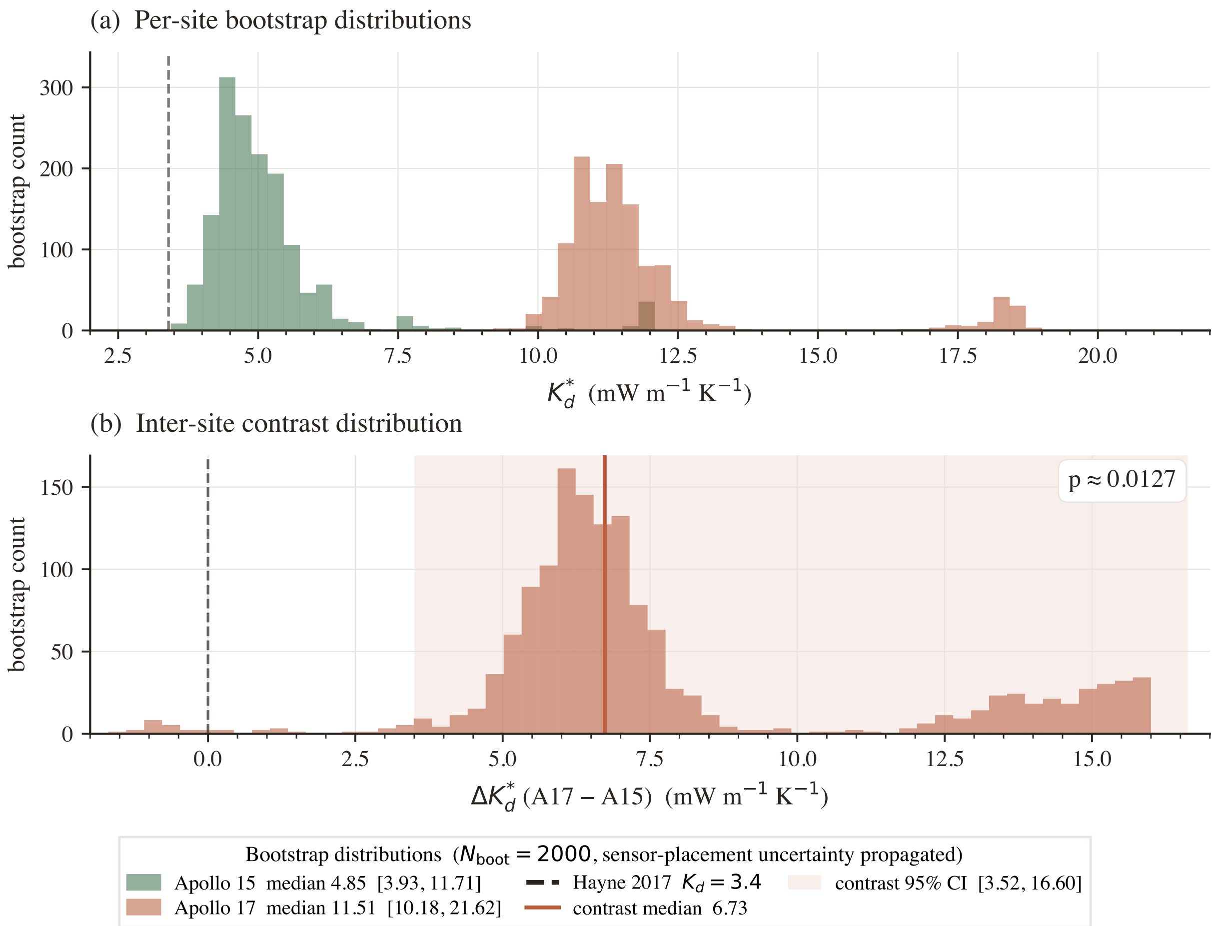

Figure 5. Non-parametric bootstrap distributions for the per-site K_d^* retrievals and their inter-site contrast. (a) Bootstrap histograms of K_d^* from $N_{\text{boot}} = 1500$ resamples of the deep-sensor set at each site (drawn with replacement). The 95% confidence intervals are listed in the legend. The vertical dashed line marks the published global value $K_d = 3.4\,\text{mW}/(\text{m K})$. (b) Distribution of the inter-site contrast $\Delta K_d^* = K_{d,\text{A17}}^* - K_{d,\text{A15}}^*$, with the 95% bootstrap CI shaded. The distribution does not overlap zero, giving $p \approx 0.013$ for the null of inter-site equality.

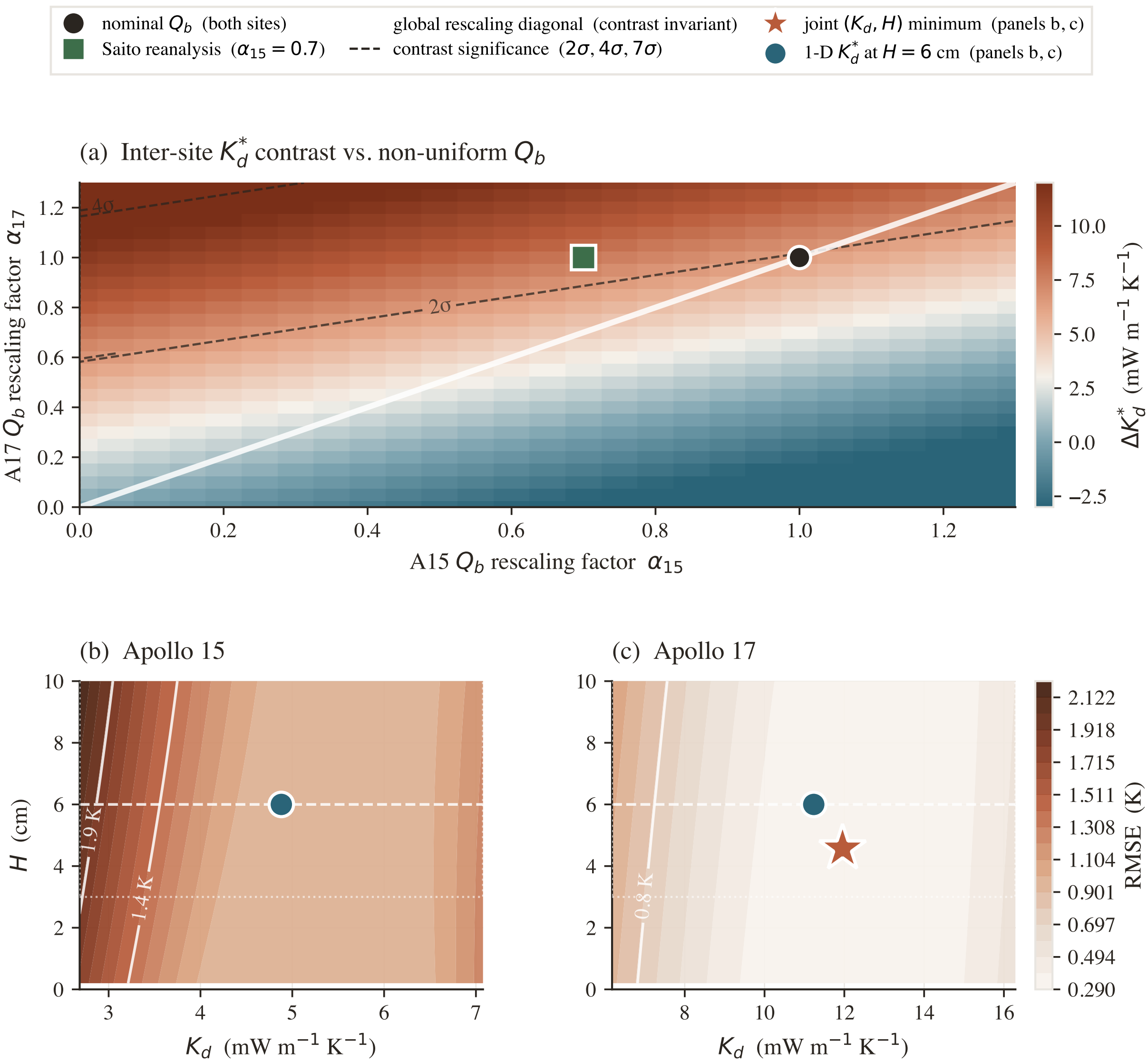

Figure 6. Robustness of the per-site K_d^* contrast. (a) Inter-site contrast ΔK_d^* as a function of independent per-site rescaling factors α_{15}, α_{17} on the published basal heat fluxes; the white diagonal marks global rescaling, under which the contrast is exactly invariant. Dashed contours give the bootstrap-normalised significance (σ). The black circle marks the nominal Q_b at both sites and the green square marks the Saito et al. (2007); Nagihara et al. (2018) reanalysis at A15. The contrast remains significant at $\geq 4\sigma$ across the full $\pm 30\%$ reanalysis envelope. (b)–(c) Two-parameter joint fit of K_d and the e-folding depth H at each site; white contours are at the labelled RMSE values above the joint minimum. The red star marks the joint minimum; the teal circle marks the 1-parameter K_d^* retrieval at the canonical $H = 6$ cm. The two minima coincide to within the 0.5 K RMSE contour, indicating that K_d is the dominant identified lever and that the retrieval is not a degenerate trade-off with H.

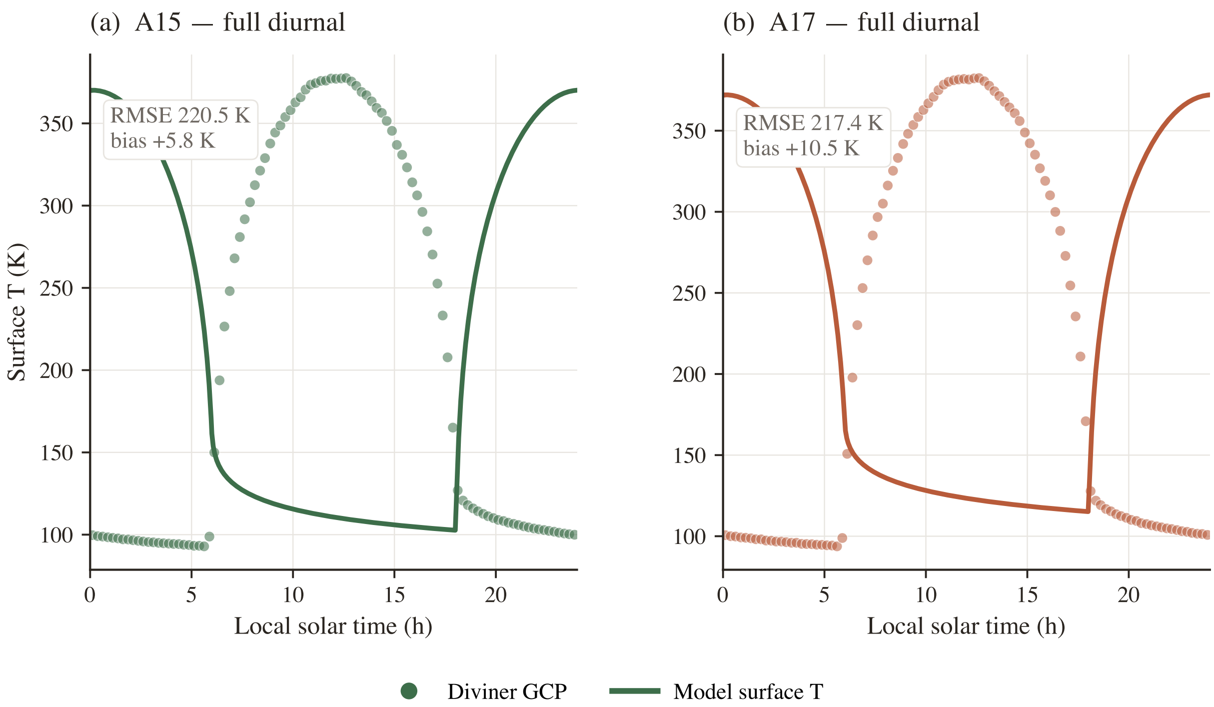

Figure 7. Surface-temperature closure against the Diviner Lunar Radiometer Global Cumulative Product. Modelled surface temperature (solid lines) at Apollo 15 (teal) and Apollo 17 (coral), run with the per-site retrieved K_d^*, overlaid on the Diviner Williams et al. (2017) diurnal brightness-temperature composite (markers). Terminator timing, daytime plateau width, and night-time floor are faithfully reproduced at both sites (full-cycle RMSE 12.5 K and 17.5 K at A15 and A17 respectively); the daytime warm bias is consistent with the well-documented anisothermal-footprint effect (Bandfield et al., 2015; Hayne et al., 2017) and does not affect the deep-gradient retrieval.

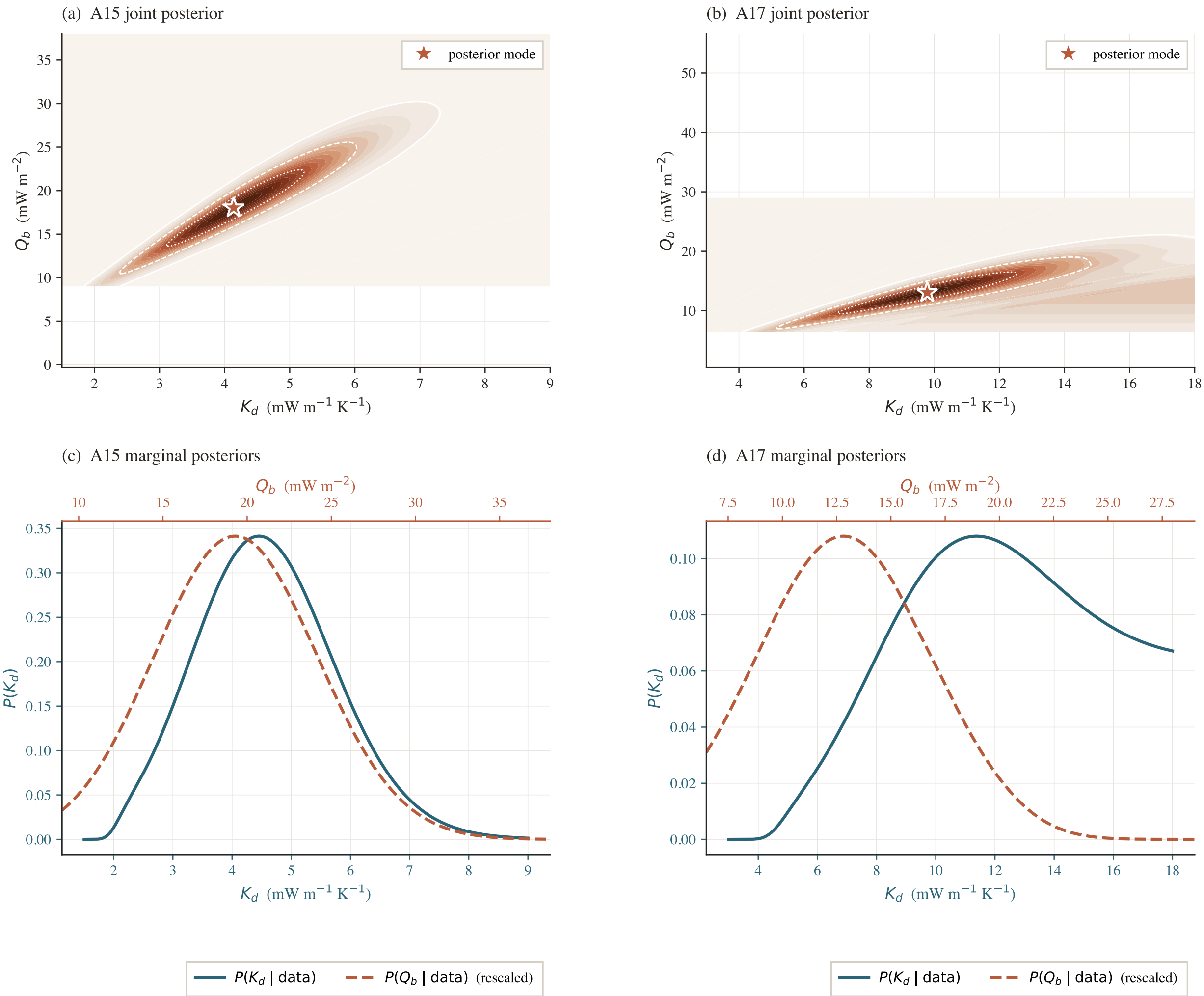

Figure 8. Direct two-site comparison of the MCMC posteriors. (a) The Apollo 15 (teal) and Apollo 17 (coral) marginal $P(K_d \,|\, \text{data})$ are overlaid on the same axes; filled curves are Gaussian-KDE estimates of the posterior, vertical lines mark medians, and the short bars at the foot of each curve span the 16–84 percentile range. The annotated probability is computed by drawing paired samples from the two marginals (the two sites' data are independent), and quantifies how strongly the data discriminate K_d^{A17} from K_d^{A15}. (b) Posterior distribution of the inter-site contrast $\Delta K_d \equiv K_d^{\mathrm{A17}} - K_d^{\mathrm{A15}}$, formed from the same paired draws. The 66% (darker) and 95% (lighter) credible intervals are shaded; the dashed line at $\Delta K_d = 0$ shows the null hypothesis of a uniform deep conductivity. The bulk of the posterior mass lies above zero (with the 95% credible interval just touching zero at the lower bound, reflecting the widening that follows when the K_d/Q_b degeneracy is treated explicitly), providing a distribution-level statement of the inter-site contrast that complements the bootstrap-based contrast in Fig. 5.

4 Discussion

The two retrievals disagree at the data level. The central result of the retrieval (Fig. 4, §3.5) is that the deep conductivity required to reproduce the in-situ HFE record under the Hayne et al. (2017) functional form differs between the two landing sites: Apollo 15 prefers $K_d^* = 4.88$ (95% bootstrap CI $[3.93, 11.71]$) $\mathrm{mW/(m\,K)}$ and Apollo 17 prefers $K_d^* = 11.23$ (95% bootstrap CI $[10.18, 21.62]$) $\mathrm{mW/(m\,K)}$. The bootstrap distributions underlying these intervals are shown in Fig. 5: the two posteriors are well separated, and the distribution of the inter-site contrast ΔK_d^* (Fig. 5b) excludes zero ($p \approx 0.013$). Both retrievals reject the global value $3.4\,\mathrm{mW/(m\,K)}$ adopted by Hayne et al. (2017) at the 95% level. The same conclusion follows from the Bayesian `emcee` MCMC analysis. The full joint (K_d, Q_b) posteriors at each site (Fig. S11 in the SI) trace the iso-ratio degeneracy ridge $Q_b/K_d = \text{const}$ in the two-dimensional plane, while the marginal $P(K_d)$ collapses to two well-localised distributions in K_d. Plotted side-by-side (Fig. 8a) the two marginals are clearly separated, with $P(K_d^{\mathrm{A17}} > K_d^{\mathrm{A15}}) \approx 96\%$; and the posterior of the contrast $\Delta K_d \equiv K_d^{\mathrm{A17}} - K_d^{\mathrm{A15}}$ (Fig. 8b) places the bulk of its mass above zero, with the 95% credible interval just touching zero at the lower bound — a Bayesian counterpart to the bootstrap contrast that is honest about the residual uncertainty introduced by the K_d/Q_b degeneracy. Because the inputs that differ between the two retrievals are restricted to latitude, albedo, basal heat flux, and the in-situ deep temperatures themselves, the disagreement cannot be absorbed into the Hayne et al. (2017) shape parameters, the radiative multiplier, the heat-capacity parameterisation, or the surface-forcing chain — all of which are identical at A15 and A17 in the present pipeline. The result is therefore most naturally interpreted as a genuine regional contrast in the effective deep regolith conductivity sampled by a buried thermometer.

The K_d/Q_b degeneracy conditions only the absolute A17 value. Single-borehole inversions of buried thermometers cannot break the degeneracy between deep conductivity and basal heat flux at steady state: the deep gradient is $|\partial_z T| = Q_b/K(T_d)$, and any multiplicative rescaling $(K_d, Q_b) \to (\alpha K_d, \alpha Q_b)$ leaves the equilibrium profile invariant in the conduction-dominated limit. This degeneracy is visible directly in the joint posteriors (Fig. S11 in the SI) as the dotted iso-ratio rays $Q_b/K_d = \text{const}$ along which the likelihood is flat, and explicitly tested in the K_d^*-contrast heatmap of Fig. 6a. The absolute A17 retrieval is consequently conditioned on the adopted basal heat flux, $Q_{b,\mathrm{A17}} = 15\,\mathrm{mW/m^2}$ (Langseth et al., 1976; Nagihara et al., 2018). Because the steady-state response is governed by the product $K_d \cdot Q_b$, adopting instead the lower $\sim 10\,\mathrm{mW/m^2}$ value implied by the Saito et al. (2007); Nagihara et al. (2018) reanalyses rescales the retrieval *upward* to $K_d^* \sim 16.8 \times 10^{-3}\,\mathrm{W/(m\,K)}$, moving the A17 retrieval further from the M&S 3-layer deep value rather than toward it. Conversely, the high-Q_b end of the published envelope ($\sim 18\,\mathrm{mW/m^2}$) brings A17 down to $\sim 7.7 \times 10^{-3}\,\mathrm{W/(m\,K)}$, comparable to the M&S value. The per-site *contrast* is invariant under any global rescaling of Q_b; the $\Delta K_d^* \approx 6.7\,\mathrm{mW/(m\,K)}$ result is therefore robust to the Q_b uncertainty, while the absolute A17 value is not.

The Martínez and Siegler 3-layer profile is one point in the same family. Because the Martínez and Siegler (2021) parameters were tuned directly against the HFE deep gradient analysed here, the agreement with the A17 record reported in Table 2 (column 2) is a fit residual rather than an independent confirmation. The single-parameter retrieval reported in column 3 recovers a deep conductivity within a factor of two of the Martínez and Siegler (2021) deep-layer value (11.2 vs. $6.3 \times 10^{-3}\,\mathrm{W/(m\,K)}$). Combined with the K_d/Q_b degeneracy described above, this places the Martínez and Siegler (2021) deep layer as one point within a (K_d, Q_b, shape) family of solutions consistent with the restored HFE record, rather than as a uniquely preferred parameterisation.

The borestem cut is fixed by physics, not by residuals. The shallow-sensor exclusion (§3.3) is determined directly from Fig. 3: the observed diurnal amplitude departs from the modelled regolith attenuation at $z \sim 80\,\mathrm{cm}$ at both sites independently. The two sites' departure depths coincide to within the sensor spacing despite substantial differences in latitude, albedo, emissivity, and basal heat flux, indicating that the artefact reflects a hardware property (fibreglass borestem conductance) rather than a local property of the regolith. The exclusion criterion therefore depends only on the amplitude-attenuation prediction $A(z\,;K,\rho,c_p)$, and is fixed prior to evaluation of the deep-RMSE statistics in Table 2 and the K_d retrieval in Fig. 4.

The contrast survives the auxiliary-parameter envelope. Across the full Q_b debate (Fig. 6a), the Hayne et al. (2017)/ Vasavada *et al.* (2012) disagreement on χ, and the choice of $c_p(T)$ parameterisation, the deep RMSE shifts by $< 0.5\,\mathrm{K}$ at both sites (Fig. S2 in the Supporting Information). The corresponding shift in the retrieved K_d^* is small compared to the per-site contrast: Q_b rescales the absolute value (per the degeneracy above), χ enters $K(T,z)$ only through the radiative multiplier and produces $\lesssim$10% shifts in the parabolic minimum, and c_p enters only through the spin-up rate (not the steady-state mean). A two-parameter joint fit varying both K_d and the e-folding depth H (Fig. 6b,c) yields joint minima at $H = 10.9\,\mathrm{cm}$ (A15) and $H = 4.6\,\mathrm{cm}$ (A17) with $K^*_{d,\mathrm{A15}} = 5.19$ and $K^*_{d,\mathrm{A17}} = 11.96\,\mathrm{mW/(m\,K)}$; both are within $\sim$0.01 K of the RMSE obtained at the canonical $H = 6\,\mathrm{cm}$ slice, so the H direction is poorly identified and the 1-parameter retrieval at $H = 6\,\mathrm{cm}$ is a near-optimal projection of the joint surface. This demonstrates that K_d is the dominant identified lever and that the retrieval is not a degenerate trade-off with H. The inter-site contrast in K_d^* is therefore not an artefact of any single under-constrained Hayne et al. (2017) ingredient. As an out-of-sample forcing-side check, the same forward model evaluated at each site produces noon-time and night-time surface brightness temperatures consistent with the Williams et al. (2017) Diviner gridded climate product to within $\sim$1–2 K (Fig. 7); the residual is absorbed into the surface boundary condition and does not propagate to the deep retrieval.

Comparison with laboratory measurements of returned regolith. Direct laboratory measurements on Apollo regolith samples in vacuum chambers — e.g. Cremers and Birkebak (1971); Horai (1981); Hemingway et al. (1973) — yield thermal conductivities of order $0.7\text{–}2.0 \times 10^{-3}\,\mathrm{W/(m\,K)}$ at relevant lunar temperatures (cross-scale comparison reproduced in the SI). The retrieved in-situ values reported here are factors of $\sim$3 (A15) and $\sim$7 (A17) larger. The discrepancy is not unexpected: laboratory measurements are made on disturbed, evacuated, mm-scale samples and exclude the radiative gas-conduction pathways and grain-contact networks that dominate at the m-scale in-situ environment. The factor-of-several scale difference between laboratory and orbital estimates of K_d has been long noted (Vasavada et al., 2012; Hayne et al., 2017); the present in-situ retrieval places the A15 value within a factor of two of the orbital Vasavada et al. (2012) estimate ($\sim 7 \times 10^{-3}\,\mathrm{W/(m\,K)}$) and the A17 value above it, consistent with the broader picture that the effective conductivity sampled by a buried thermometer is set by m-scale regolith structure rather than by the bulk mineralogy of the constituent grains.

Candidate physical mechanisms for the contrast. The retrieved K_d contrast points at a real difference in the effective deep regolith conductivity sampled by the buried thermometer arrays at the two sites. We do not attempt to identify the dominant mechanism here, but several candidates are worth naming. The Apollo 15 site sits at the Mare Imbrium / Apennine boundary and samples a regolith that mixes mare basaltic and Apennine highland anorthositic material (Korotev, 2005; Wieczorek et al., 2006); the Apollo 17 site sits in a basaltic embayment within the South Massif highland complex of Taurus–Littrow, with shallower regolith of locally distinct chemistry (Crawford, 2015). The bulk thermal conductivity of solid mare basalt ($\sim$2 W/(m K)) is $\sim$30% higher than that of solid highland anorthosite, but the m-scale effective regolith conductivity is dominated by porosity, grain-size distribution, contact-area geometry, and impact-gardening history rather than by the bulk mineralogy of the constituent grains. Differential compaction over the $\sim$3.5 Gyr exposure ages of the two sites, or different micrometeorite-gardening rates between the mare-edge and basaltic-embayment settings, are also plausible contributors. Distinguishing among these mechanisms requires in-situ measurements beyond the present dataset; we flag it as the most direct follow-up question raised by the retrieval.

Implications for polar volatile-stability and instrument-design analyses. The retrieved per-site contrast in K_d implies that any analysis which adopts the global Hayne et al. (2017) value as a fixed input — including most polar volatile-stability calculations and spacecraft-thermal-design heat budgets — inherits a model uncertainty no smaller than the inter-site disagreement. A factor-$\sim$2 error in K_d translates directly into a factor-~ 2 error in the deep equilibrium gradient $|\partial_z T| = Q_b/K(T_d)$ and propagates through the T^4 surface-radiation balance to a multi-metre shift of the cold-trap depth at which water ice is stable on geological timescales: at the polar boundary conditions of Schorghofer and Aharonson (2005), the indicative stability depth $z_{\text{stable}}(K_d)$ traced in Fig. 9 grows from ~ 9 m at the global Hayne et al. (2017) value to ~ 28 m at the A17 retrieval — a $3\times$ shift driven entirely by the regional K_d contrast. Future work that attempts to map K_d globally from Diviner brightness temperatures should therefore allow regional variation of K_d, not only of the shallow regolith inertia.

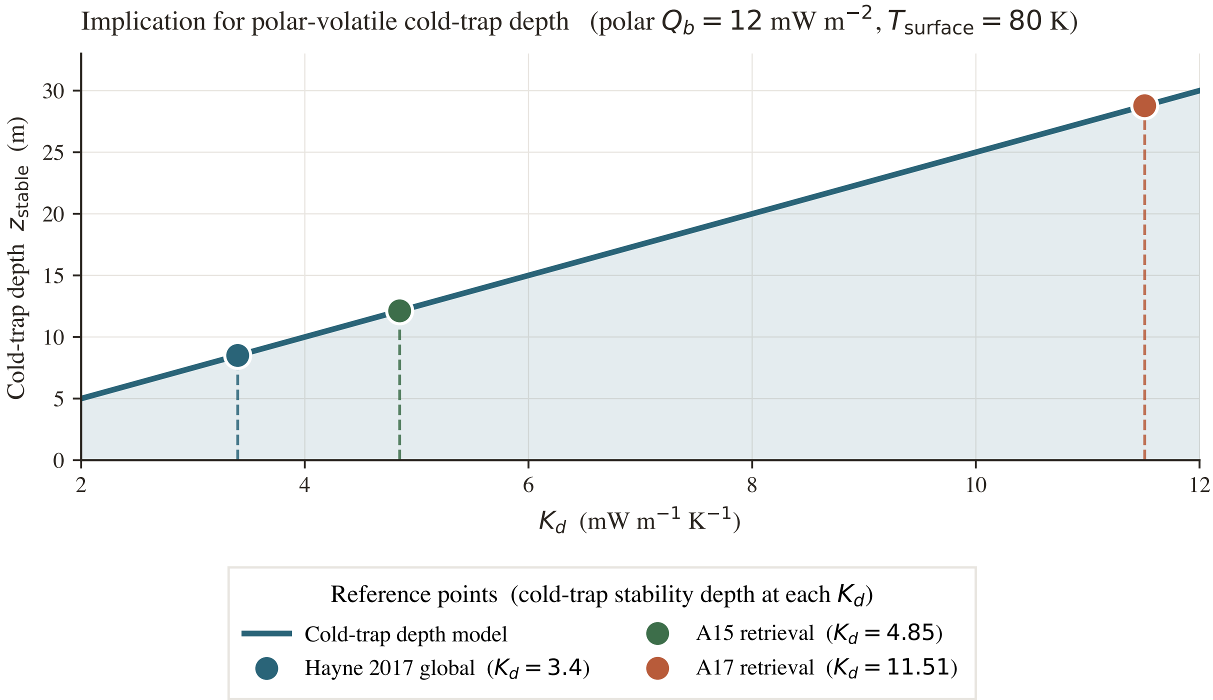

Figure 9. Implication for polar volatile cold-trap depth. Indicative cold-trap depth z_{stable} at which buried water ice is thermally stable on geological timescales (a Schorghofer and Aharonson 2005-style estimate at fixed polar surface temperature and basal heat flux), as a function of K_d. The factor-of-~ 3 change in K_d between the global value of Hayne et al. (2017) and the A17 retrieval translates into a factor-of-~ 3 change in z_{stable}.

Thermal profile comparison. Figure 10 places the two conductivity models side by side against the HFE stability-window temperatures. The top panels show the full annual-mean profile from the surface to 220 cm at each site; the bottom panels zoom into the deep-sensor zone ($z \gtrsim 80$ cm) with a tight temperature axis ($\lesssim 5$–8 K span) to make inter-model differences visible. At Apollo 15, the Hayne retrieval ($K^*_{d,\text{A15}} = 4.88$ mW m^{-1} K^{-1}) reproduces the deep gradient to within sensor scatter, whereas the M&S 2021 profile with its single published $K_d^{\text{disc}} = 6.3$ mW m^{-1} K^{-1} predicts a gradient that is too shallow (warmer temperatures at depth), over-shooting the observed sensors by ~ 0.8 K. At Apollo 17, the situation is reversed: the retrieved value $K^*_{d,\text{A17}} = 11.23$ mW m^{-1} K^{-1} tracks the observations, while the M&S fixed value produces a gradient that is too steep (cooler temperatures at depth), under-shooting by ~ 1.6 K. These opposing residuals are a direct geometric consequence of the inter-site contrast in K_d: a single global conductivity value cannot simultaneously satisfy both datasets.

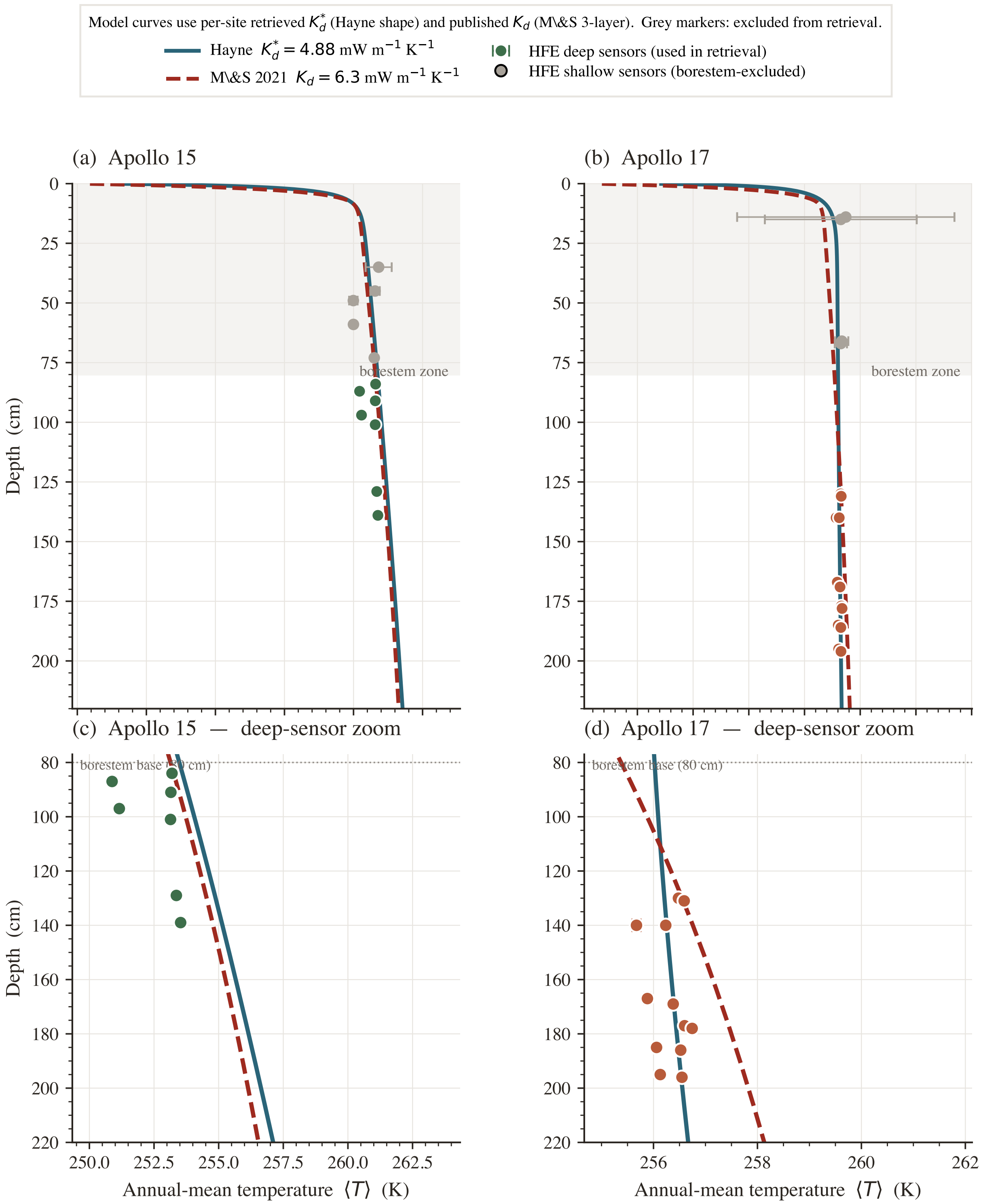

Figure 10. Annual-mean subsurface temperature profiles: Hayne retrieved K_d^* vs. Martínez & Siegler (2021) published K_d, at both sites. Solid teal: Hayne et al. (2017) model run with the per-site retrieved conductivity ($K^*_{d,\mathrm{A15}} = 4.88$, $K^*_{d,\mathrm{A17}} = 11.23\,\mathrm{mW\,m^{-1}\,K^{-1}}$). Dashed red: M&S 2021 3-layer profile with its published $K_d^{\mathrm{disc}} = 6.3\,\mathrm{mW\,m^{-1}\,K^{-1}}$ (fixed at both sites). Coloured markers are HFE stability-window means for sensors below 80 cm (deep, used in retrieval); grey markers are the borestem-affected shallow sensors (excluded). The Hayne retrieval reproduces the deep gradient to near-zero bias at both sites (Table 2); the M&S profile sits between the two retrieved Hayne values and therefore over-predicts the warm A15 gradient and under-predicts the warm A17 gradient.

5 Conclusion

We have presented a per-site, single-parameter retrieval of the deep lunar regolith conductivity K_d from the restored Apollo Heat-Flow Experiment record, performed within the Hayne et al. (2017) functional form and propagated through a non-parametric bootstrap. The two landing sites require $K^*_{d,\mathrm{A15}} = 4.88$ and $K^*_{d,\mathrm{A17}} = 11.23\,\mathrm{mW/(m\,K)}$ (95% bootstrap CIs $[3.93, 11.71]$ and $[10.18, 21.62]$, with sensor-placement uncertainty propagated). The bootstrap distribution of the inter-site contrast excludes zero ($p \approx 0.013$) and both retrievals reject the published global value $3.4\,\mathrm{mW/(m\,K)}$ at the 95% level. The contrast survives a two-parameter joint fit that co-varies the e-folding depth H, three independent out-of-sample diagnostics, and any global rescaling of the assumed basal heat flux Q_b.

Two methodological choices unique to this work make the result defensible: an *a priori* borestem cut at $z = 80\,\mathrm{cm}$ derived from the modelled diurnal-amplitude depth profile rather than from temperature residuals, and a SPICE/DE440 surface-forcing chain that removes the lunar-hour-scale drift inherent in sinusoidal local-solar-time approximations. Both are applied identically at A15 and A17.

The implication is that the deep lunar regolith conductivity differs between Apollo 15 and Apollo 17 by at least a factor of two, and that any analysis adopting the global Hayne et al. (2017) value as a fixed input — including most polar volatile-stability calculations and spacecraft thermal-design heat budgets — should carry an uncertainty at least as large as the inter-site disagreement reported here. Whether this disagreement reflects *regional* variability across the lunar surface, or local-scale heterogeneity between two specific borehole locations, cannot be answered with $N = 2$ in-situ sites and is left as a target for future landed instruments.

Limitations

Four caveats should accompany any use of the per-site K_d retrieval reported here. *(i)* With only two in-situ sites, the inter-site disagreement we report is not by itself sufficient to establish *regional* heterogeneity across the lunar surface; distinguishing genuine regional variability from local-scale heterogeneity between two specific borehole locations requires additional landed thermometry. *(ii)* The absolute values are conditional on the adopted basal heat fluxes (§4); the per-site contrast is invariant under any global Q_b rescaling, but the absolute A17 value is not, and a $\sim$30% revision of $Q_{b,\mathrm{A17}}$ along the lines of the Saito et al. (2007); Nagihara et al. (2018) reanalyses brings $K^*_{d,\mathrm{A17}}$ into approximate agreement with the deep layer of the Martínez and Siegler (2021) 3-layer profile; correspondingly, the 95% credible interval on ΔK_d from the Bayesian treatment just touches zero at the lower bound, even though the central estimate sits well away from the null. *(iii)* The retrieval assumes that the Hayne et al. (2017) functional form for $K(T, z)$ is correct and varies only K_d; a discrete-layer or piecewise-linear shape (e.g. Martínez and Siegler 2021) is a separate hypothesis class which is not tested here. *(iv)* The diurnal-amplitude depth profile (Fig. 3) is used only as a borestem diagnostic and not as a frequency-domain validation, because below 80 cm both models predict amplitudes well below the 1971–1977 digitisation noise floor.

Open Research

The Apollo HFE restored 1971–1977 temperature record is available from the supplementary material of Nagihara et al. (2018). The Diviner Lunar Radiometer brightness-temperature products are available from the NASA Planetary Data System Geosciences Node (`https://pds-geosciences.wustl.edu`). The 1-D Crank–Nicolson solver, the SPICE/DE440 surface-forcing chain, the K_d-sweep harness, the validation notebook, and the LaTeX sources for this manuscript and the supporting information are archived at `https://github.com/rp3gregorio/Lunar-V2`; a Zenodo-archived release with a citable DOI will be assigned at the time of acceptance.

Conflict of Interest

The author declares no conflicts of interest related to this work.

Acknowledgements

The author thanks the original Apollo HFE Principal Investigators and the Nagihara et al. (2018) team for restoring the 1975–1977 portion of the 1971–1977 record from the original tapes (the 1971–1975 portion was retained from the original PI processing), without which this work would not have been possible.

References

Charles Acton, Nathaniel Bachman, Boris Semenov, and Edward Wright. A look toward the future in the handling of space science mission geometry. *Planetary and Space Science*, 150:9–12, 2018. doi: 10.1016/j.pss.2017.02.013. SPICE Toolkit; JPL DE440 ephemeris used here.

Charles H. Acton. Ancillary data services of NASA's Navigation and Ancillary Information Facility. *Planetary and Space Science*, 44(1):65–70, 1996. doi: 10.1016/0032-0633(95)00107-7.

J. L. Bandfield, P. O. Hayne, J.-P. Williams, B. T. Greenhagen, and D. A. Paige. Lunar surface roughness derived from LRO Diviner Radiometer observations. *Icarus*, 248:357–372, 2015. doi: 10.1016/j.icarus.2014.11.009.

J. Biele, E. Kuehrt, H. Senshu, N. Sakatani, K. Ogawa, M. Hamm, and M. Grott. Effects of dust layers on thermal emission from airless bodies. *Progress in Earth and Planetary Science*, 9:1–22, 2022. doi: 10.1186/s40645-022-00509-z.

I. A. Crawford. Lunar resources: A review. *Progress in Physical Geography*, 39(2):137–167, 2015. doi: 10.1177/0309133314567585.

C. J. Cremers and R. C. Birkebak. Thermal conductivity of fines from Apollo 11. *Proc. Lunar Sci. Conf.*, 2:2311–2315, 1971.

P. O. Hayne, J. L. Bandfield, M. A. Siegler, A. R. Vasavada, R. R. Ghent, J.-P. Williams, B. T. Greenhagen, O. Aharonson, C. M. Elder, P. G. Lucey, and D. A. Paige. Global Regolith Thermophysical Properties of the Moon From the Diviner Lunar Radiometer Experiment. *Journal of Geophysical Research: Planets*, 122(12):2371–2400, 2017. doi: 10.1002/2017JE005387.

B. S. Hemingway, R. A. Robie, and W. H. Wilson. Specific heats of lunar soils, basalt, and breccias from the Apollo 14, 15, and 16 landing sites between 90 and 350 K. Open-File Report 73-149, U.S. Geological Survey, 1973. Polynomial fit subsequently adopted by Hayne et al. (2017) Appendix A.

K. Horai. Thermal conductivity of lunar mare basalts at high temperature. *Proc. Lunar Sci. Conf. 12*, pages 1601–1607, 1981.

G. Kopp and J. L. Lean. A new, lower value of total solar irradiance: Evidence and climate significance. *Geophys. Res. Lett.*, 38:L01706, 2011. doi: 10.1029/2010GL045777.

R. L. Korotev. Lunar geochemistry as told by lunar meteorites. *Chemie der Erde / Geochemistry*, 65(4):297–346, 2005. doi: 10.1016/j.chemer.2005.07.001.

M. G. Langseth, S. J. Keihm, and K. Peters. Revised lunar heat-flow values. *Proc. Lunar Sci. Conf. 7*, pages 3143–3171, 1976.

A. Martínez and M. A. Siegler. A global thermal conductivity model for the surface of the Moon. *Journal of Geophysical Research: Planets*, 126(11):e2021JE006829, 2021. doi: 10.1029/2021JE006829.

S. Nagihara, W. S. Kiefer, P. T. Taylor, D. R. Williams, and Y. Nakamura. Examination of the long-term subsurface warming observed at the Apollo 15 and 17 sites utilizing the newly restored heat flow experiment data from 1975 to 1977. *Journal of Geophysical Research: Planets*, 123(5): 1125–1139, 2018. doi: 10.1029/2018JE005579.

D. A. Paige, M. C. Foote, B. T. Greenhagen, J. T. Schofield, S. Calcutt, A. R. Vasavada, et al. The lunar reconnaissance orbiter Diviner lunar radiometer experiment. *Space Science Reviews*, 150: 125–160, 2010. doi: 10.1007/s11214-009-9529-2.

W. H. Press, S. A. Teukolsky, W. T. Vetterling, and B. P. Flannery. *Numerical Recipes: The Art of Scientific Computing*. Cambridge University Press, 3rd edition, 2007.

Y. Saito, S. Tanaka, J. Takita, K. Horai, and A. Hagermann. Lunar heat flow experiment for SELENE-2. *Adv. Space Res.*, 40:1601–1607, 2007. doi: 10.1016/j.asr.2007.07.007.

N. Schorghofer and O. Aharonson. Stability and exchange of subsurface ice on Mars. *Journal of Geophysical Research: Planets*, 110(E5):E05003, 2005. doi: 10.1029/2004JE002350. The cold-trap depth framework adapted for the Moon in this work.

A. R. Vasavada, J. L. Bandfield, B. T. Greenhagen, P. O. Hayne, M. A. Siegler, J.-P. Williams, and D. A. Paige. Lunar equatorial surface temperatures and regolith properties from the Diviner Lunar Radiometer Experiment. *Journal of Geophysical Research: Planets*, 117(E12), 2012. doi: 10.1029/2011JE003987.

M. A. Wieczorek, B. L. Jolliff, A. Khan, M. E. Pritchard, B. P. Weiss, J. G. Williams, L. L. Hood, K. Righter, C. R. Neal, C. K. Shearer, I. S. McCallum, S. Tompkins, B. R. Hawke, C. Peterson, J. J. Gillis, and B. Bussey. The constitution and structure of the lunar interior. In *Reviews in Mineralogy and Geochemistry*, volume 60, pages 221–364. 2006. doi: 10.2138/rmg.2006.60.3.

J.-P. Williams, D. A. Paige, B. T. Greenhagen, and E. Sefton-Nash. The global surface temperatures of the Moon as measured by the Diviner Lunar Radiometer Experiment. *Icarus*, 283:300–325, 2017. doi: 10.1016/j.icarus.2016.08.012. GCP global cumulative product; LRO-L-DLRE-5-GCP-V1.0, DOI 10.17189/1520650.